Zhenyu Liao

Curriculum Vitae

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Male, Chinese, born on 28/08/1992.

Education

2019	**Ph.D.**	Mathematics and Informatics	L2S, CentraleSupélec, Paris-Saclay University, France.
2016	**M.Sc.**	Signal and Image Processing	Paris-Saclay University, France.
2014	**B.E.**	Optical & Electronic Information	Huazhong University of Science and Technology, China.

Professional Experience

- 2021–present: **Associate Professor**, School of Electronic Information and Communications, Huazhong University of Science and Technology.
- 2020–2021: **Postdoctoral Scholar**, International Computer Science Institute (ICSI) and Department of Statistics, University of California, Berkeley, hosted by Michael Mahoney.

Honors and Awards

- 2026: Top AC of NeurIPS 2026
- 2026: The Springer Nature Editor of Distinction – *Author Service Award 2026* for Statistics and Computing.
- 2025: IEEE Senior Member.
- 2025: Recipient of the MATRIX-Simons Travel Grant, MATRIX, Creswick, Australia.
- 2025: **CRM-Simons Visiting Scholar**, Centre de recherches mathématiques (CRM), Université de Montréal, for the thematic program Mathematical Foundations of Data Science. *(Unable to attend because of visa issues.)*
- 2024: **ANR-LabEx-CIMI Visiting Professor**, Centre International de Mathématiques et Informatique de Toulouse (CIMI), Toulouse, France.
- 2023: Recipient of a Talent Program of Hubei Province, China.
- 2021: Recipient of the Wuhan Youth Talent Program, Wuhan, China.
- 2021: Recipient of the East Lake Youth Talent Program Fellowship of HUST, Wuhan, China.
- 2019: Second Prize, ED STIC Doctoral Prize, Paris-Saclay University, France.

Selected Publications

Books

1. Romain Couillet and **Zhenyu Liao**. *Random Matrix Methods for Machine Learning*. Cambridge University Press, 2022. DOI: 10.1017/9781009128490.

Papers in conference proceedings

1. Chengmei Niu, Sachin Garg, Michal Derezinski, and **Zhenyu Liao**. Debiasing Random Oblique Projections for Subsampled OLS and Fast CUR in High Dimensions. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. 2026.
2. Yanlei Liu and **Zhenyu Liao**. Dual-Attention Convolution Experts for Sparse Tensor Completion. In: *Machine Learning and Knowledge Discovery in Databases. Research Track (**ECML PKDD**)*. Springer Nature Switzerland, Sept. 2027, pp.450–467. https://link.springer.com/chapter/10.1007/978-3-032-37670-1_26.
3. Mohamed Chiheb Yaakoubi, Cosme Louart, Malik Tiomoko, and **Zhenyu Liao**. Characterization of Gaussian Universality Breakdown in High-Dimensional Empirical Risk Minimization. In: *Proceedings of the 43rd International Conference on Machine Learning (**ICML**)*. Vol. 306. PMLR, July 2026, pp.143332–143358. https://proceedings.mlr.press/v306/yaakoubi26a.html.
4. Jiayu Bai, Danchen Yu, **Zhenyu Liao**, Tianqi Hou, Feng Zhou, Robert C. Qiu, and Zenan Ling. Diving into Kronecker Adapters: Component Design Matters. In: *Proceedings of the 43rd International Conference on Machine Learning (**ICML**)*. Vol. 306. PMLR, July 2026, pp.5479–5501. https://proceedings.mlr.press/v306/bai26l.html.
5. Yue Xu, Shuo Zhang, Gui Zhou, and **Zhenyu Liao**. Optimal Large-scale Finite-horizon Optimization via Random Matrix Theory. In: *2026 34th European Signal Processing Conference (**EUSIPCO**)*. 2026.
6. Yue Xu and **Zhenyu Liao**. An Improved Convergence Analysis of Gossip Methods for Large Random Graphs. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. 2026, pp.22562–22566. DOI: 10.1109/ICASSP55912.2026.11461477.

7. Yessin Moahker, Malik Tiomoko, Cosme Louart, and **Zhenyu Liao**. A Random Matrix Perspective of Echo State Networks: From Precise Bias-Variance Characterization to Optimal Regularization. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. 2026, pp.20716–20720. DOI: 10.1109/ICASSP55912.2026.11464627.
8. Jiuqi Wei, **Zhenyu Liao**, Ruoyu Han, Quanqing Xu, Chuanhui Yang, and Themis Palpanas. TaCo: Data-adaptive and Query-aware Subspace Collision for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (**SIGMOD**)* **4**(3) (May 2026). DOI: 10.1145/3802118.
9. Chengmei Niu, **Zhenyu Liao**, Zenan Ling, and Michael W. Mahoney. Fundamental Bias in Inverting Random Sampling Matrices with Application to Sub-sampled Newton. In: *Proceedings of the 42nd International Conference on Machine Learning (**ICML**)*. Vol. 267. PMLR, July 2025, pp.46649–46692. https://proceedings.mlr.press/v267/niu25c.html.
10. Xiaoyi Mai and **Zhenyu Liao**. The Breakdown of Gaussian Universality in Classification of High-dimensional Linear Factor Mixtures. In: *International Conference on Learning Representations (**ICLR**)*. 2025. https://openreview.net/forum?id=UrKbn51HjA.
11. Jiuqi Wei, Xiaodong Lee, **Zhenyu Liao**, Themis Palpanas, and Botao Peng. Subspace Collision: An Efficient and Accurate Framework for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (**SIGMOD**)* **3**(1) (Feb. 2025). DOI: 10.1145/3709729.
12. Wei Yang, Zhengyu Wang, Xiaoyi Mai, Zenan Ling, Robert Caiming Qiu, and **Zhenyu Liao**. Inconsistency of ESPRIT DoA Estimation for Large Arrays and a Correction via RMT. In: *2024 32nd European Signal Processing Conference (**EUSIPCO**)*. Aug. 2024, pp.2722–2726. DOI: 10.23919/EUSIPCO63174.2024.10715081.
13. Zenan Ling, Longbo Li, Zhanbo Feng, Yixuan Zhang, Feng Zhou, Robert C. Qiu, and **Zhenyu Liao**. Deep Equilibrium Models Are Almost Equivalent to Not-so-deep Explicit Models for High-dimensional Gaussian Mixtures. In: *Proceedings of the 41st International Conference on Machine Learning (**ICML**)*. Vol. 235. PMLR, July 2024, pp.30585–30609. https://proceedings.mlr.press/v235/ling24a.html.
14. Lingyu Gu, Yongqi Du, Yuan Zhang, Di Xie, Shiliang Pu, Robert Qiu, and **Zhenyu Liao**. "Lossless" Compression of Deep Neural Networks: A High-dimensional Neural Tangent Kernel Approach. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 35. Curran Associates, Inc., 2022, pp.3774–3787. https://proceedings.neurips.cc/paper_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html.
15. Hafiz Tiomoko Ali, **Zhenyu Liao**, and Romain Couillet. Random matrices in service of ML footprint: ternary random features with no performance loss. In: *International Conference on Learning Representations (**ICLR**)*. 2022. https://openreview.net/forum?id=qwULHx9zld.
16. **Zhenyu Liao** and Michael W Mahoney. Hessian Eigenspectra of More Realistic Nonlinear Models. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 34. Curran Associates, Inc., 2021, pp.20104–20117. https://papers.nips.cc/paper/2021/hash/a7d8ae4569120b5bec12e7b6e9648b86-Abstract.html.
17. Michal Derezinski, **Zhenyu Liao**, Edgar Dobriban, and Michael Mahoney. Sparse sketches with small inversion bias. In: *Proceedings of Thirty Fourth Conference on Learning Theory (**COLT**)*. Vol. 134. PMLR, Aug. 2021, pp.1467–1510. https://proceedings.mlr.press/v134/derezinski21a.html.
18. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. Sparse Quantized Spectral Clustering. In: *International Conference on Learning Representations (**ICLR**)*. 2021. https://openreview.net/forum?id=pBqLS-7KYAF.
19. Fanghui Liu, **Zhenyu Liao**, and Johan Suykens. Kernel Regression in High Dimension: Refined Analysis beyond Double Descent. In: *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (**AISTATS**)*. Vol. 130. PMLR, Apr. 2021, pp.649–657. http://proceedings.mlr.press/v130/liu21b.html.
20. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A Random Matrix Analysis of Random Fourier Features: Beyond the Gaussian Kernel, A Precise Phase Transition, and the Corresponding Double Descent. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., 2020, pp.13939–13950. https://papers.nips.cc/paper/2020/hash/a03fa30821986dff10fc66647c84c9c3-Abstract.html.
21. Michal Derezinski, Feynman T Liang, **Zhenyu Liao**, and Michael W. Mahoney. Precise expressions for random projections: Low-rank approximation and randomized Newton. In: *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 33. Curran Associates, Inc., 2020, pp.18272–18283. https://papers.nips.cc/paper/2020/hash/d40d35b3063c11244fbf38e9b55074be-Abstract.html.
22. Xiaoyi Mai, **Zhenyu Liao**, and Romain Couillet. A Large Scale Analysis of Logistic Regression: Asymptotic Performance and New Insights. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE. May 2019, pp.3357–3361. DOI: 10.1109/ICASSP.2019.8683376.

23. Romain Couillet, **Zhenyu Liao**, and Xiaoyi Mai. Classification Asymptotics in the Random Matrix Regime. In: *The 26th European Signal Processing Conference (**EUSIPCO**)*. IEEE. Sept. 2018, pp.1875–1879. DOI: 10.23919/EUSIPCO.2018.8553034.
24. **Zhenyu Liao** and Romain Couillet. The Dynamics of Learning: A Random Matrix Approach. In: *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, July 2018, pp.3072–3081. http://proceedings.mlr.press/v80/liao18b.html.
25. **Zhenyu Liao** and Romain Couillet. On the Spectrum of Random Features Maps of High Dimensional Data. In: *Proceedings of the 35th International Conference on Machine Learning (**ICML**)*. Vol. 80. PMLR, July 2018, pp.3063–3071. http://proceedings.mlr.press/v80/liao18a.html.
26. **Zhenyu Liao** and Romain Couillet. Random Matrices Meet Machine Learning: A Large Dimensional Analysis of LS-SVM. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*. IEEE. Mar. 2017, pp.2397–2401. DOI: 10.1109/ICASSP.2017.7952586.

Journal papers

1. **Zhenyu Liao** and Michael W. Mahoney. Random Matrix Theory for Deep Learning: Beyond Eigenvalues of Linear Models. *IEEE Signal Processing Magazine* **43**(2) (2026), 93–106. DOI: 10.1109/MSP.2025.3618012.
2. Zhengyu Wang, Wei Yang, Xiaoyi Mai, Zenan Ling, **Zhenyu Liao**, and Robert C. Qiu. A Large-Dimensional Analysis of ESPRIT DoA Estimation: Inconsistency and a Correction via RMT. *IEEE Transactions on Signal Processing* **74** (2026), 2290–2303. DOI: 10.1109/TSP.2026.3683673.
3. Junchao Lin, Zenan Ling, Zhanbo Feng, Jingwen Xu, Minxuan Liao, Feng Zhou, Tianqi Hou, **Zhenyu Liao**, and Robert C. Qiu. IGNN-Solver: A Graph Neural Solver for Implicit Graph Neural Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026), 1–18. DOI: 10.1109/TPAMI.2026.3729092.
4. Zhanbo Feng, Yuanjie Wang, Jie Li, Fan Yang, Jiong Lou, Tiebin Mi, Robert. C. Qiu, and **Zhenyu Liao**. Robust and Communication-Efficient Federated Domain Adaptation via Random Features. *IEEE Transactions on Knowledge and Data Engineering* **37**(3) (2025), 1411–1424. DOI: 10.1109/TKDE.2024.3510296.
5. Yacine Chitour, **Zhenyu Liao**, and Romain Couillet. A geometric approach of gradient descent algorithms in linear neural networks. *Mathematical Control and Related Fields* **13**(3) (2023), 918–945. DOI: 10.3934/mcrf.2022021.
6. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A random matrix analysis of random Fourier features: beyond the Gaussian kernel, a precise phase transition, and the corresponding double descent. *Journal of Statistical Mechanics: Theory and Experiment* **2021**(12) (Dec. 2021), 124006. DOI: 10.1088/1742-5468/ac3a77.
7. **Zhenyu Liao** and Romain Couillet. A Large Dimensional Analysis of Least Squares Support Vector Machines. *IEEE Transactions on Signal Processing* **67**(4) (Feb. 2019), 1065–1074. DOI: 10.1109/TSP.2018.2889954.
8. Cosme Louart, **Zhenyu Liao**, and Romain Couillet. A Random Matrix Approach to Neural Networks. *The Annals of Applied Probability* **28**(2) (Apr. 2018), 1190–1248. DOI: 10.1214/17-AAP1328.

Research Grants

1. 2026–2029: **PI**, National Natural Science Foundation of China General Program, *Compression of Large-scale Transformer-based Models with Random Matrix Methods: From Scaling Law to Dynamic Adaptive Model Compression* (NSFC-12571561), CNY 440K, with Cosme Louart as co-PI.
2. 2025–2030: **Co-PI**, National Key R&D Program of China (Youth Scientist Project), *Foundations of Machine Learning for Understanding Large Models* (2025YFA1018600), CNY 1M/3M. **PI:** Jun Shu, Xi'an Jiaotong University.
3. 2026–2028: **PI**, HUST Basic Research Support Program (Mathematics Track), *Theory and Methods for Large-scale Machine Learning via RMT* (2025BRSXB0004), CNY 380K.
4. 2024–2026: **PI**, Guangdong Key Lab of Mathematical Foundations for Artificial Intelligence Open Fund, *Generalization Theory for Transformer-based Models via Random Matrix Methods* (OFA00003), CNY 100K, with Jeff Yao as co-PI.
5. 2023–2025: **PI**, National Natural Science Foundation of China Youth Program, *Fundamental Limits of Pruning Deep Neural Network Models via Random Matrix Methods* (NSFC-62206101), CNY 300K.
6. 2022–2025: **Contributor**, National Natural Science Foundation of China grant, *Mathematical Foundations for Future Communications (Information Theory)* (NSFC-12141107), CNY 3M. PI: Robert C. Qiu.
7. 2021–2022: **PI**, CCF-Hikvision Open Fund, *Random Matrix Theory and Information Bottleneck for Neural Network Compression* (20210008), CNY 280K, with Kai Wan as co-PI.
8. 2018–2021: **Contributor**, NSF Research Grant, *Combining Stochastics and Numerics for Improved Scalable Matrix Computations* (NSF-1815054), USD 500K. PI: Michael W. Mahoney.

9. 2018–2021: **Contributor**, Programme d'Investissements d'avenir, *GSTATS IDEX DataScience Chair*, Université Grenoble Alpes, EUR 300K. PI: Romain Couillet.
10. 2014–2017: **Contributor**, French National Research Agency, *Random Matrix Theory for Large Dimensional Graphs* (ANR-14-CE28-0006), EUR 300K. PI: Romain Couillet.

Professional Services

- Conference service for ICML, NeurIPS, ICLR, AISTATS, IJCNN, and AAAI.
- Associate Editor for Springer Statistics and Computing.
- Guest editor of the Special Topic "Random Matrix Theory and Its Applications" in SCIENCE CHINA Mathematics

Organization of Scientific Activities

1. Co-organizer, with Jianfeng Yao, Fan Yang, Jiang Hu, Dandan Jiang, and Shurong Zheng, of the 2027 Annual Meeting on Random Matrix Theory and Applications, January 14–17, 2027, Sanya, China.
2. Co-organizer, with Zhigang Bao, Benoît Collins, Dong Yao, Jianfeng Yao, and Shurong Zheng, the TMRC-FRMTA Workshop, August 16–22, 2026, Yunnan, China
3. **Sponsorship Co-chair** of the 2026 IEEE/CIC International Conference on Communications in China (ICCC), August 7–9, 2026.
4. Co-organizer, with Jun Shu, of the *Modern Machine Learning Theory and Methods* session at CSML 2026, August 7–9, 2026, Shanghai, China.
5. Co-organizer, with Fanghui Liu and Lei Wu, of the TM10 session *Theory Guides Practice for Large Models* at CSIAM-BDAI 2026, July 17–19, 2026, Xiamen, China.
6. Co-organizer, with Rishabh Dudeja, Junjie Ma, and Arian Maleki, of the Universality and Dynamics in High-Dimensional Learning and Inference Workshop at ISIT 2026, July 3, 2026, Guangzhou, China.
7. Co-organizer, with Dirk Slock, of the Special Session on *Large-Dimensional Structures and Methods for Signal Processing, Communication, and Machine Learning* at ICASSP 2026, May 2026, Barcelona, Spain.
8. Co-organizer, with Pengkun Yang and Cong Fang, of the 2025 Mathematical and Statistical Foundations of Foundation Models Workshop, July 2025, Changbai Mountain, China.
9. Co-organizer of the Founding Conference of the Random Matrix Theory and Applications Branch of CAAS, January 2025, Changchun, China.
10. **General Co-chair** of the 10th EAI International Conference on IoT as a Service (EAI IoTaaS 2024), Wuhan, China.
11. Co-organizer of the 1st High-dimensional Learning Dynamics (HiLD) Workshop at ICML 2023, with Mihai Nica, Courtney Paquette, Elliot Paquette, Andrew Saxe, and René Vidal.
12. Co-organizer of the 2022 HUST–Paris-Saclay Joint Workshop on Math for Data Science with Yacine Chitour; more than 40,000 online attendees.

Peer Reviewing

- Invited external examiner for a Ph.D. thesis, The University of Queensland, Australia (2026).
- Invited external examiner for a Ph.D. thesis, Tsinghua University, China (2021).
- **Research grants**: European Research Council (ERC), Natural Sciences and Engineering Research Council of Canada (NSERC), Israel Science Foundation (ISF), National Natural Science Foundation of China (NSFC).
- **Conferences**: ICLR, ICML, NeurIPS, SODA, AISTATS, AAAI, ECAI, IEEE-ICASSP, IEEE-CAMSAP.
- **Journals**:
  - Mathematics, statistics, and data science: SIAM Review (SIREV), Foundations of Computational Mathematics (FoCM), Physical Review X (PRX), IEEE Transactions on Information Theory (TIT), Journal of Mathematical Physics, Springer Statistics and Computing (STCO), SIAM Journal on Scientific Computing (SISC), Random Matrices: Theory and Applications (RMTA), Latin American Journal of Probability and Mathematical Statistics (ALEA), Annals of Applied Probability (AAP), Electronic Communications in Probability (ECP), Electronic Journal of Statistics (EJS), Journal of Applied Mathematics and Computing (JAMC), AIMS Mathematics.
  - Machine learning and AI: Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE-TPAMI), IEEE Transactions on Dependable and Secure Computing (TDSC), IEEE Signal Processing Letters (SPL), IEEE Transactions on Signal Processing (IEEE-TSP), IEEE Transactions on Multimedia (IEEE-TMM), Engineering Applications of Artificial Intelligence (EAAI), Frontiers of Computer Science, Neural Networks (NEUNET), IEEE Transactions on Neural Networks and Learning Systems (IEEE-TNNLS), IEEE Transactions on Industrial Informatics (IEEE-TII), Transactions on Machine Learning Research (TMLR), Pattern Recognition (PR), Neural Processing Letters (NPL).